HAPPIEST HEALTH SUMMIT 2026, BENGALURU · DIGITAL TWIN CHALLENGE: TYPE 2 DIABETES

GlucoTwin

A personal digital twin that sees the next two hours of glucose

Record + CGM + sleep and activity → a 2-hour forecast → early warnings with reasons → 7 plans on her own twin.

18.3%
lower 2-hour error than persistence, on 45 real adults

12%
lower 1-hour error once the health record sets the prior

105 min
median warning before a spike; 109 of 254 warned 60+ min ahead

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Phase 1 · prototype & code submission · October 2026

Research prototype, not a medical device.
All patient data is synthetic or open.

Doctors see glucose after the fact. Nothing lets them look ahead.

A diabetologist's clinic, between two visits

Ananya Rao, 54, has Type 2 diabetes: HbA1c 8.1%, BMI 29.4, on metformin. She is seen every three months.

Her CGM writes 288 readings a day. The doctor gets a 14-day PDF, a separate lab report, and a few minutes.

"Which meal is doing this? Would an earlier dinner help her, or hurt her?" The data cannot answer.

By the time HbA1c moves, the spikes have already happened. Advice stays generic: "cut rice, walk more".

THREE GAPS IN TODAY'S CARE

- **Record and sensor data never meet**
- **Reports look back; nothing looks ahead**
- **No way to test a plan on this patient**

36.3% reach the glycaemic target

Only 7.7% of people with diabetes in India reach HbA1c, blood pressure and LDL targets together.

The missing piece is foresight, not more data.

THE STAKES, NATIONALLY

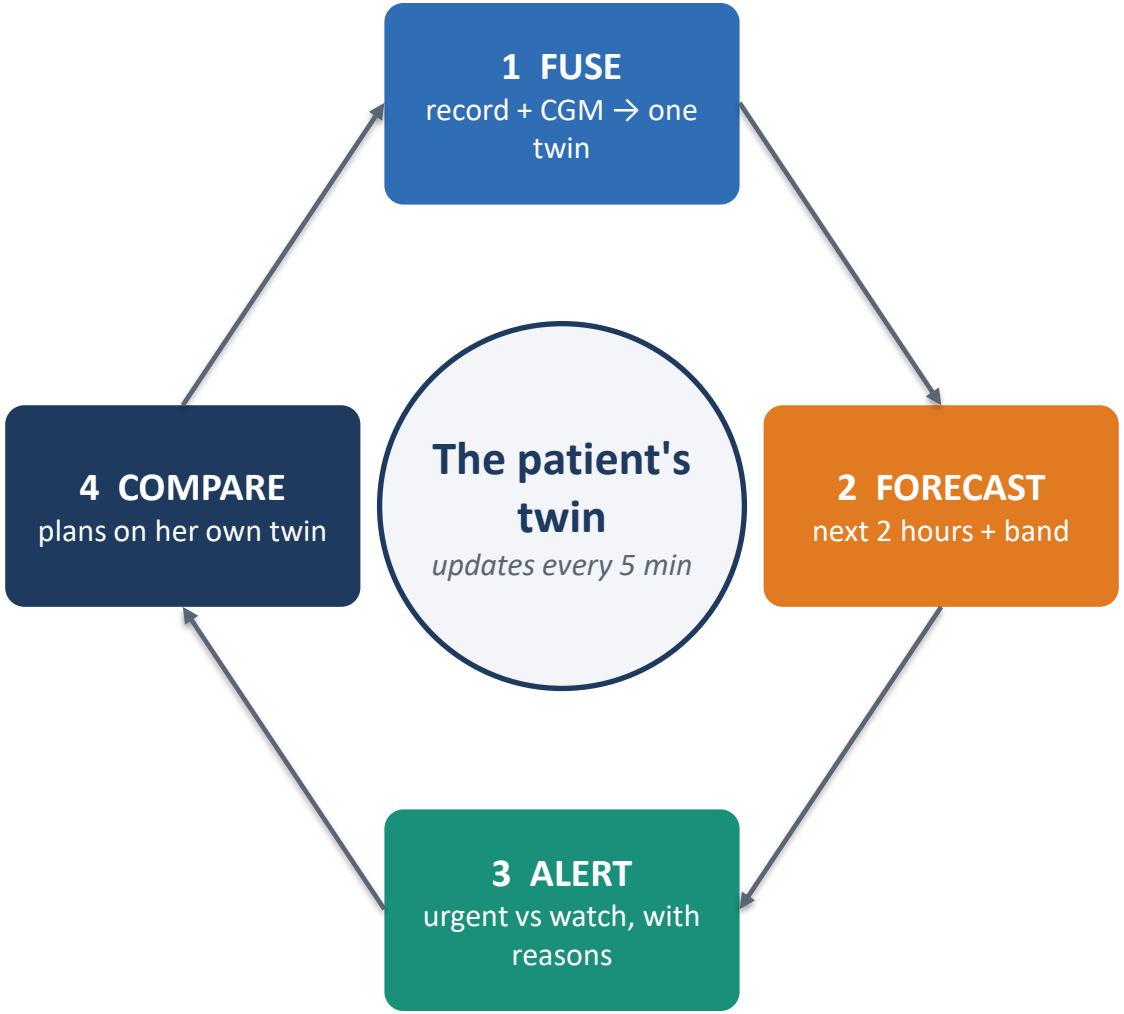
101M adults with diabetes, plus 136M with prediabetes (ICMR-INDIAB)

156.7M adults with diabetes by 2050, up 75% from 2024 (IDF)

43% of Indian adults with diabetes are undiagnosed (IDF 2025)

Sources: ICMR-INDIAB-17 (Lancet Diabetes Endocrinol 2023); ICMR-INDIAB-13 (Lancet Diabetes Endocrinol 2022); IDF Diabetes Atlas 11th ed. (2025), India. Patient is simulated (GlucoTwin ward, GT-07); the clinic scene is illustrative.

GlucoTwin turns data into the next two hours, for one patient



Doctors already get data. What is missing is what happens next.

1 FUSE

The FHIR R4 record (labs, BMI, medicines, genetic marker) sets day-0 knobs; CGM, sleep, HRV and walks move the live state.

−12%
1-hour error, day 0

2 FORECAST

Physiology curve from meals, walks, dawn rise, sleep and HRV, with a 10–90% band fitted on the twin's own past errors.

−18.3%
2-hour error

3 ALERT

Spike risk includes her habitual next meal; thresholds are chosen on validation days. Only Urgent interrupts; every alert lists reasons.

105 min
median warning

4 COMPARE

Seven lifestyle plans, incl. a weekly fasting day, on her highest-risk meal. Never a dose; “Simulation, not advice”.

+3.6 pts
time in range

Sources: GlucoTwin results. Error figures on CGMacros (PhysioNet, 45 adults); alert figures on the simulated stress test (generator differs from the twin); plan figure on the simulated ward, days 10–16.

Every line of the brief, met and measured

The brief asks for	GlucoTwin delivers	Evidence
A chronic condition prevalent in India	Type 2 diabetes and prediabetes, with Indian subgroups and IFCT 2017 foods	89.8M Indian adults (IDF 2025)
Static record: demographics, diagnoses, labs, genetic markers	FHIR R4 bundle: age, sex, HbA1c trail, fasting glucose, HOMA-IR, BMI, medicines, simulated TCF7L2 marker	Record cuts 1-hour error 12%
Dynamic wearable / IoT time series	CGM every 5 min, meals, walks, heart rate, sleep and HRV (they shift insulin sensitivity), steps	45 real adults + 11 simulated
A model that predicts an adverse event ahead of time	Glucose at 30, 60 and 120 min with a 10–90% band; spike and low alerts with reasons	Median 105-min warning; 109 of 254 ≥ 1 h
A conceptual doctor dashboard	Working dashboard, 7 screens, per-stream consent gate and a printable pre-visit summary	Runs live in any browser
Synthetic, open or anonymised data only	Synthetic ward (seed 42) + open CGMacros (PhysioNet); no real identities anywhere	DPDP / HIPAA sandbox rules met

Two streams fused · an adverse event predicted · a doctor's dashboard · open and synthetic data only

Brief: Happiest Health Digital Twin Challenge 2026. Evidence figures come from the result pages that follow (CGMacros, PhysioNet; simulated stress test; IDF Diabetes Atlas 2025).

What the twin already produces

REAL PROTOTYPE OUTPUT · SIMULATED WARD (11 PATIENTS) + REAL CGMACROS DATA (45 ADULTS)

Patient	Fact from the twin	What the doctor gets
Ananya Rao (GT-07), simulated	After a 67 g carb lunch, the 2-hour forecast rises past 180 mg/dL; spike risk crosses 60%	Urgent alert with reasons: the meal, glucose vs her fasting baseline, time of day
Split lunch, 8 Type 2 patients	Vikram +3.6 and Kavita +2.9 points of time in range; Leela −5.4, Fatima −4.1, Suresh −2.9	A personal plan table, not a one-size rule
All 8 Type 2 patients, simulated	Personal 60-min error below the group prior for every one (gain 0.5–3.7 mg/dL)	A twin fitted to each person in 5 days
3 real CGMacros adults, held out	LightGBM residual cuts their 2-hour error from 46.5 to 40.8 mg/dL	Offline today; candidate for the live curve

Honest check: on real CGMacros data, spike alerts caught 38.9% of spikes at 63.8% precision (median lead 35 min) and 7 of 17 lows at 39% precision. The habit prior is not yet on the real-data path; Indian pilot data is the fix.

```
$ cat real_eval.json
participants          45
forecasts/horizon     11,101
rmse_120  twin        31.8
rmse_120  persist     38.9
clarke_AB_30          0.998
band_cover_60         0.794
$ node --test  # model+data
tests 118  pass 118  fail 0
```

BUILT · habitual-meal anticipation

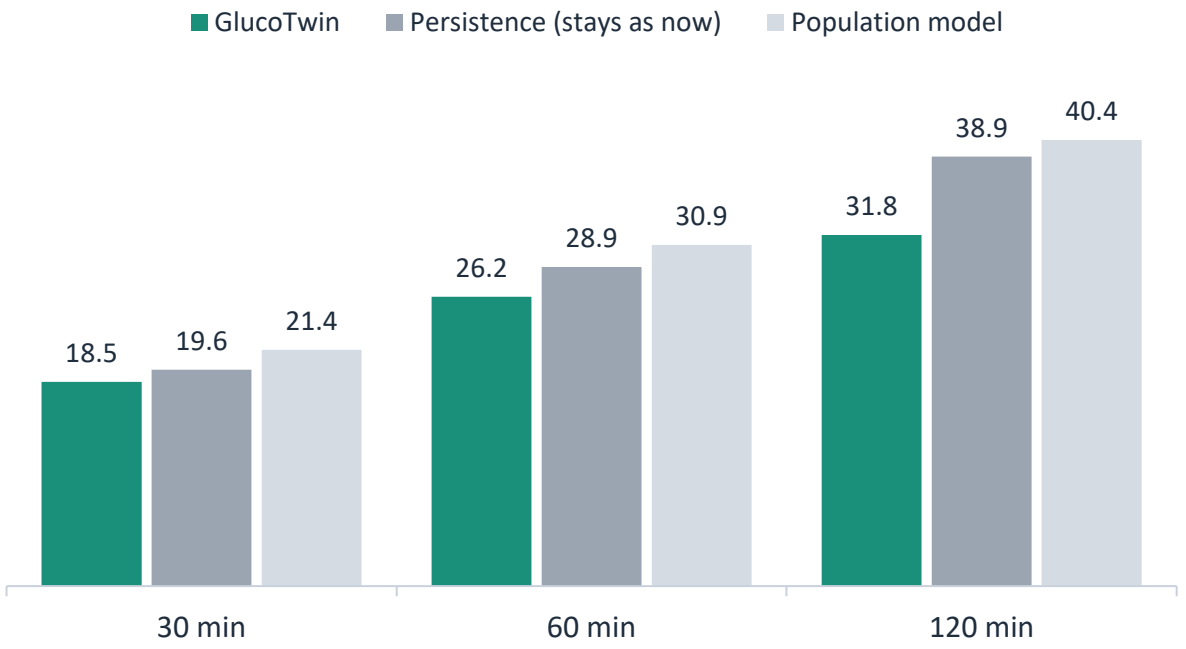
Spike risk includes her usual next meal: 109 of 254 spikes warned 60+ min ahead (was 7 of 244), 33 a full 2 hours ahead.

Then: the LightGBM residual joins the live curve once the pilot review signs it off.

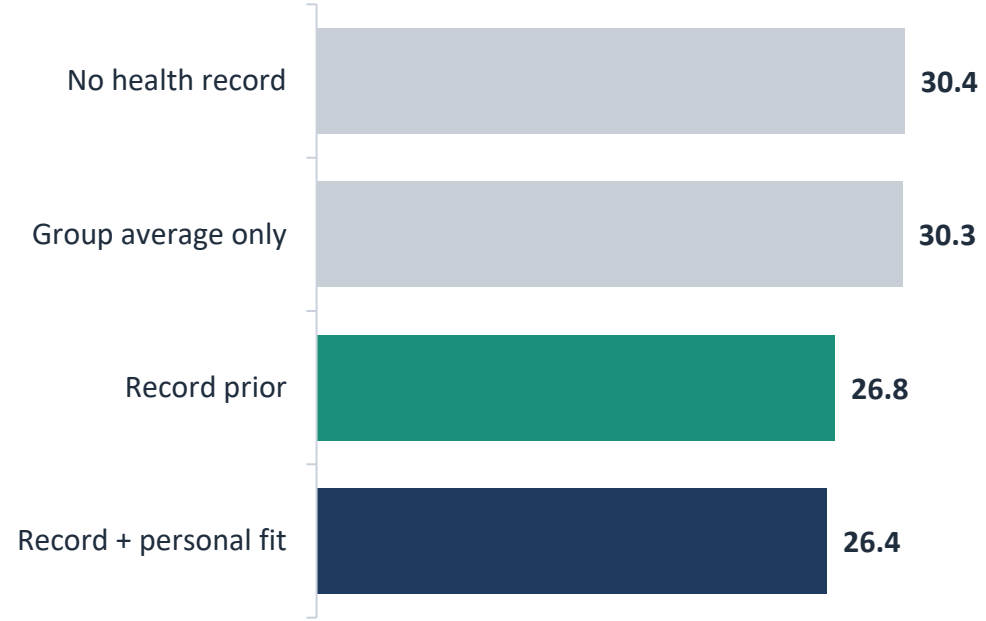
Sources: GlucoTwin outputs — reports/sim_eval.json (simulated ward, seed 42); src/lib/glucotwin/real/real_eval.json (CGMacros, doi:10.13026/3z8q-x658); model and data test run, 17 Oct 2026.

On 45 real adults, the twin beats the baseline where it matters

FORECAST ERROR, RMSE IN MG/DL (LOWER IS BETTER)



1-HOUR ERROR BY WHAT THE TWIN KNOWS AT START



18.3%
lower 2-hour error than persistence (31.8 vs 38.9 mg/dL)

12%
lower 1-hour error when the record sets the prior (30.4 → 26.8)

99.8%
of 30-minute forecasts in Clarke error-grid zones A and B

8–10%
lower 1-hour error in every group: Type 2, prediabetes, healthy

CGMacros (PhysioNet, CC BY-NC-SA 4.0): 45 adults (15 healthy, 16 prediabetes, 14 Type 2). Knobs fit on days 1–4, band on days 5–6, test from day 7; ~11,100 forecasts per horizon. Ablation: leave-one-out ridge prior, 11,192 forecasts per arm.

Warnings arrive early, and plans are personal

SPIKE ALERTS ON THE STRESS TEST

61.0%
of spikes warned (95% CI 54.9–66.8%)

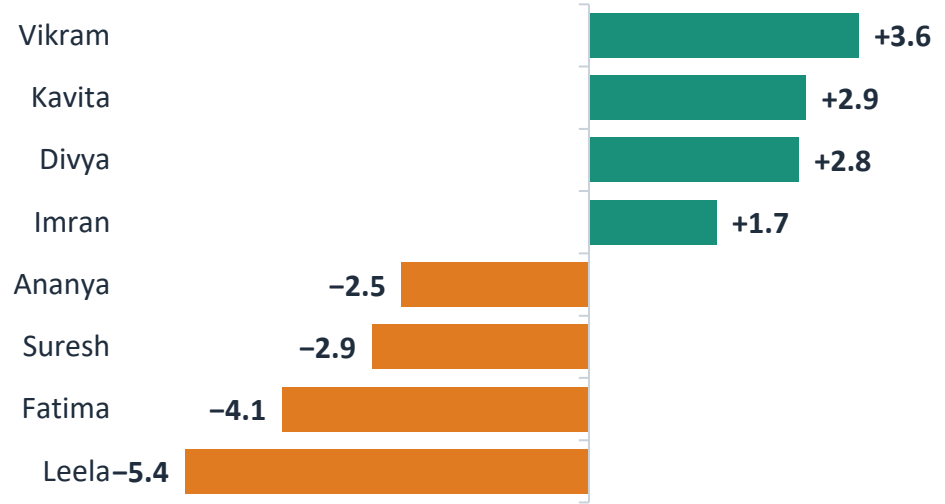
66.9%
of spike alerts were right (95% CI 61.4–72.0%)

105 min
median warning before the spike (IQR 50–115)

0.75
false alerts per patient-day

Only Urgent alerts interrupt, and **83.5%** of them were right. A Watch is upgraded to Urgent if the curve gets worse.

SAME PLAN, DIFFERENT TWINS: SPLIT LUNCH, CHANGE IN TIME IN RANGE (PTS)



It helps 4 of 8 Type 2 twins and hurts the other 4: a one-size rule would be wrong for half. Lifestyle plans only, never a dose.

WHY IT MATTERS

+5 pts
of time in range is clinically meaningful (international consensus, 2019)

10 pts
of time in range track about 0.8% of HbA1c (Vigersky & McMahon, 2019)

105 min
of warning leaves time to change the meal itself, not only walk after it

Alert figures: simulated stress test (generator differs from the twin), 11 patients, 254 spikes, Wilson 95% CIs. Plan figures: simulated ward, days 10–16, time in range 70–180 mg/dL. Battelino et al., Diabetes Care 2019; Vigersky & McMahon, Diabetes Technol Ther 2019.

Level with the best simple alarm on F1, with a third of its false alarms

SAME 254 SPIKES, SAME ALERT RULES, SAME SCORER

Method	Uses	Catches	Right	False / day	Median warning
GlucoTwin (risk ≥ 0.8)	**Record + CGM + meals + sleep**	**61.0%**	**66.9%**	**0.75**	**105 min**
Rise-rate alarm	CGM only	24.8%	78.8%	0.13	5 min
Trend projection, 60 min	CGM only	89.4%	49.5%	2.11	25 min
Trend projection, 120 min	CGM only	81.5%	34.5%	3.86	45 min
Big-meal rule, ≥ 40 g carbs	Meal log only	56.7%	58.5%	0.77	30 min

Catches = share of spikes warned in the 2 hours before. Right = share of alerts followed by a spike.

0.638

F1, level with the 60-min trend projection (0.637), which raises 2.8× the false alerts

3–5×

fewer false alerts than CGM trend projections (0.75 vs 2.1–3.9 a day)

2.5×

the spikes a CGM rise-rate alarm catches, and 100 min earlier

Warning time: median 105 min. 109 of 254 spikes were warned 60+ min ahead and 33 a full 2 hours, because spike risk includes the habitual next meal. CGM trend projections warn at most 45 min ahead, with 3–5× the false alarms.

Risk numbers: when the twin showed a spike risk of 95% or more, a spike followed 95.6% of the time (2,544 steps). Brier skill 0.61.

Simulated stress test (mismatched generator, seed 42): 11 patients × 12 scored days, 254 spikes. Every method uses the same events, the same alert hygiene and the same scorer. scripts/alert-proof.ts → docs/ALERT_PROOF.md. CGMacros comparison not yet run.

Built around the doctor who reviews the patient

PRIMARY

Diabetologist / physician

- Reviews CGM and labs between visits
- Wants: who to look at now, why, and what to try
- Works at a desktop; sees reasons, not black-box scores
- Decides; the twin never doses

SECONDARY

Diabetes educator / care team

- Explains the chosen plan to the patient
- Follows up Urgent alerts and plan adherence
- Needs plain reasons and simple plan tables

DATA OWNER

The patient

- Wears a CGM; logs meals and walks
- Consents per stream (DPDP-style, live in the dashboard)
- Gets advice from the doctor, never from the twin

CONTEXT THAT SHAPES THE DESIGN

62.3%

of daily energy in India comes from carbohydrate → a meal model built on IFCT 2017 Indian foods

4

Indian Type 2 subgroups (SIDD, IROD, CIRDD, MARD) → an approximate cold-start prior

93 crore

ABHA accounts with 104 crore linked records → the record arrives via ABDM

288

CGM readings a day per patient → the doctor needs triage, not more charts

Sources: Anjana et al., Nature Medicine 2025 (62.3% carbohydrate energy); docs/CLUSTERS.md (approximate subgroup rule, not INSPIRED); PIB ABDM fact sheet, 6 Jul 2026; CGM cadence of one reading per 5 minutes.

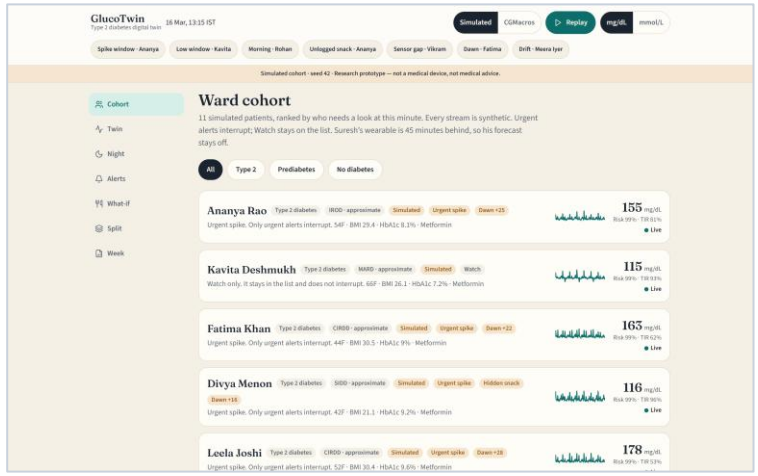
The doctor's quarter: before and after

Doctor's quarter	Data arrives	Prepare review	Spot risk	Choose a plan	Between visits	Next visit
BEFORE (today)	14-day CGM PDF and a separate lab report	Reads charts by hand; no link to the record	Spikes seen days after they happened	Generic advice: cut rice, walk more	No signal until the next visit	HbA1c moved or not; cause unknown
WITH GLUCOTWIN	Record + CGM fused into one twin from day 0	Cohort ranked by urgency on one screen	Alert a median 105 min before a spike, with reasons	Seven plans compared on her own twin	Drift check flags when the twin stops fitting	One-click pre-visit summary for her chart

Biggest shift: from reading the past to testing the next two hours, on this patient, before advising her.

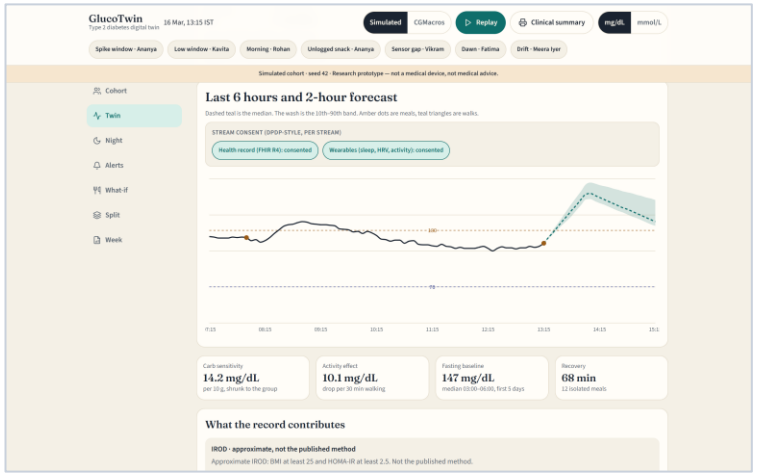
'Before' is an illustrative clinic workflow. 'After' lists features running in the prototype today. Time saved per review will be measured in the pilot, not claimed here.

Three screens a doctor uses between visits



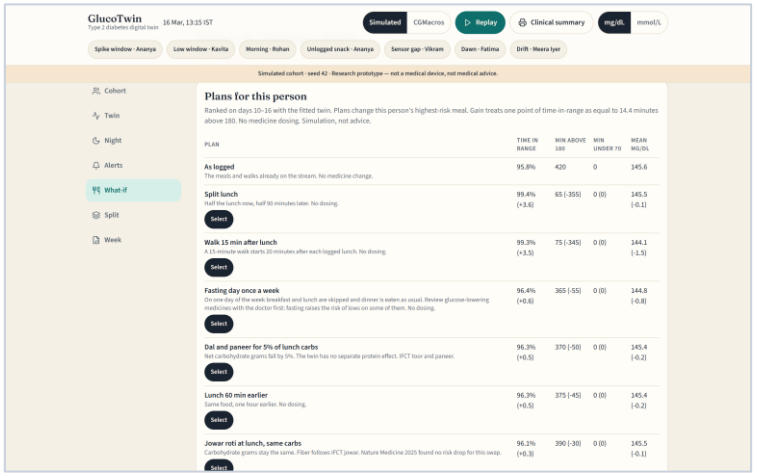
COHORT TRIAGE

Patients ranked by urgency. Only Urgent interrupts; Watch stays on the list.



TWIN FORECAST

2-hour forecast with a 10–90% conformal band and per-stream consent toggles.



7-PLAN WHAT-IF

Seven lifestyle plans on her own twin, with deltas. Never a dose.

ONE CLICK: PRE-VISIT CLINICAL SUMMARY

A printable sheet for the patient's chart: twin parameters, 7-day time in range, the chosen lifestyle plan and recent Urgent alerts.

Withdraw the record and the twin drops its prior and medicines but keeps its CGM fit; withdraw wearables and sleep and walks drop out. The band widens on screen.

Screenshots of the running prototype (simulated ward, seed 42), 16 Oct 2026 (docs/presentation/screens). The pre-visit summary prints from the Clinical summary button on any patient screen.

Others report the past. GlucoTwin tests the next two hours.

	Fuses record + CGM	Looks ahead	Personal fitted model	What-if plans	Indian foods & subgroups	Shows uncertainty
CGM vendor reports (AGP)	○ sensor only	◐ short low alarms, some devices	○	○	○ not described	○
Diabetes logging & coaching apps	◐ manual entry	○ not described	◐ varies	◐ tips, not simulated	◐ some	○
Hospital EHR / HMIS	◐ record, rarely CGM	○	○	○	○	○
GlucoTwin	**● FHIR R4 + CGM**	**● 30 / 60 / 120 min**	**● 5-day fit**	**● 7 plans**	**● IFCT + 4 subgroups**	**● 10–90% band**

● does it ◐ partly / manual ○ not described publicly

NEW: PLAN-ON-TWIN

Every lifestyle plan is tested on this patient's fitted twin before it is advised.

Split lunch helped 4 Type 2 patients and hurt 4. A report of the past cannot show that.

Explainable: every forecast splits into meal, walk, dawn rise and residual.

Honest: shows a 10–90% band and its real coverage (0.79 at one hour against a 0.80 target).

Indian by design: IFCT foods and subgroup priors, not Western food tables.

Category-level summary from public product descriptions as of Oct 2026; individual products vary. “Not described” ≠ absent. Band coverage from real_eval.json (60-min, CGMacros).

Physiology explains the curve; ML learns what it misses

PHYSIOLOGY

TypeScript twin core, live

- Meal response from carbs, fiber and fat
- Walks, dawn rise, sleep + HRV, fading residual
- 5 personal knobs, kept only if they beat the prior
- Conformal 10–90% band from its own errors
- Spike and low probability → alerts

Every curve is explainable

MACHINE LEARNING

Python, offline

- Ridge prior: labs, BMI, age, sex → day-0 knobs (−12% error)
- LightGBM learns the physiology residual
- Subgroup rule sets a cold-start prior
- Drift monitor flags when the band stops fitting
- Python ↔ TypeScript parity on 300 cases

Learns only what physiology misses

DOCTOR

Reviews and decides

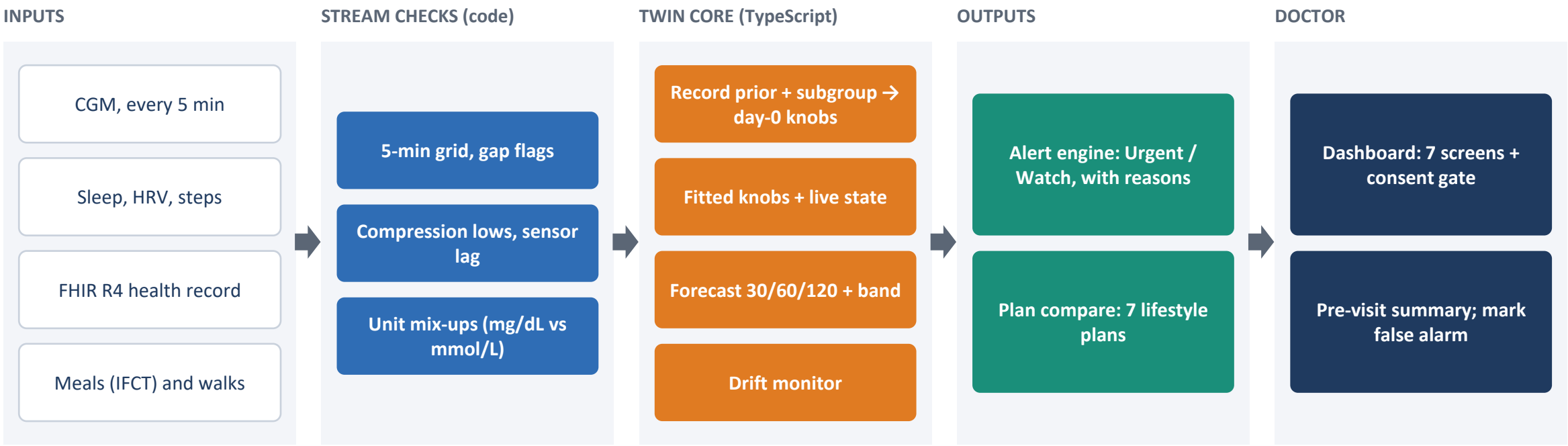
- Works from a cohort ranked by urgency
- Picks a lifestyle plan, or none
- Acknowledges or marks a false alarm
- Sees the data source on every screen
- Never sees a dose suggestion

Accountability stays with the doctor

Why not a pure black box: open data has only 45 real people with CGM and logged meals. A physiology core fitted to each person in five days generalises, explains itself, and leaves ML to correct what it misses: LightGBM cut held-out 2-hour error from 47.7 to 43.2 mg/dL.

Sources: GlucoTwin design (src/lib/glucotwin/physics.ts, engine.ts; ml/evaluate_cgmacros.py). LightGBM held-out figures from real_eval.json (3 participants held out). Parity checked by ml/parity_test.py.

How data flows, and where the twin acts



OFFLINE (Python): CGMacros → ridge prior + LightGBM residual → results the app reads

STACK

TypeScript · React 19 · TanStack Start · Recharts · Python (numpy, pandas, scikit-learn, LightGBM) · FHIR R4 · Vercel

TODAY (HONEST STATUS): RUNNING END TO END

11 simulated patients + 45 real CGMacros adults.
118/118 model tests; 16.3 µs per reading.

NEXT

ABHA-consented FHIR pull · ABDM consent artefacts · event queue sharded by patient · LightGBM on the live curve after review

Sources: GlucoTwin architecture as built (docs/architecture). Benchmark 17 Oct 2026 on one laptop thread. Items under NEXT are planned, not built. The app deploys as serverless functions (Nitro on Vercel).

Guardrails, scale and cost

Sensor noise & gaps

Stream checks flag gaps, compression lows and lag before the twin uses a reading

Alarm fatigue

Two tiers, hysteresis, thresholds chosen on validation days; 0.75 false alerts a day

Over-trust & dosing

No dose, no diagnosis; plans labelled “Simulation, not advice”

Privacy (DPDP)

Synthetic and open data only; per-stream consent gate live in the dashboard

Population shift

CGMacros is US adults; the pilot refits on Indian patients and meals

Rare lows

Stress test: 6 of 16 lows warned, all correct (blind guard). CGMacros: 7 of 17, 39% precise

SCALE (MEASURED)

- 16.3 µs per reading on one laptop core: update, 2-hour forecast and spike risk
- 1.7 KB of state per patient: a million patients fit in 1.7 GB
- O(1) update and stateless workers: shard by patient ID
- No CGM? Eight fingersticks a day rebuild the curve at 20.3 mg/dL error (simulated)
- 118/118 model tests; Python ↔ TypeScript parity on 300 cases

MODEL COMPUTE PER PATIENT (MEASURED)

≈ 0.14 CPU-seconds / month

288 readings × 30 days × 16.3 µs. One cloud vCPU covers the model for millions of patients, so the sensor, storage and hosting, not the twin, set the cost.

At 1 million patients: ≈ 39 CPU-hours of model compute a month, 3,300 readings a second, 1.7 GB of twin state.

Hosting and sensor prices are not yet costed.

Benchmark 17 Oct 2026: 3,000 synthetic patients × 288 readings, Node 22, Intel i5-11400H, one thread, compute only (no network or database I/O). Fingerstick figure: interpolation error on the simulated ward. Lows: 6 of 16 on the stress test (blind drug guard), 7 of 17 on CGMacros.

Pilot plan, impact metrics and roadmap

NOW → DEC 2026

Prototype

- Twin running: 11 simulated + 45 real adults
- Built: habit-aware alerts, consent gate, pre-visit summary

NEXT

One-clinic pilot

- 30 adults with Type 2 diabetes on CGM, 12 weeks
- Indian meals via IFCT; doctor reviews forecasts

THEN

City rollout

- More clinics via ABDM, with ABHA consent
- Hypertension twin on the same core

HOW WE'LL KNOW IT WORKS (lead metric: time in range)

Time in range

points gained vs the pilot's baseline weeks; +5 is clinically meaningful

Spikes warned early

% warned ≥60 min ahead (stress test today: 42.9%, median 105 min)

False alerts

per patient-day; today 0.75, keep below 1

Review time

minutes per patient review, before vs after

ADOPTION & SUSTAINABILITY

Plugs in: ABDM exchanges records as FHIR R4, the format the twin already reads; 93 crore ABHA accounts and 104 crore linked records exist today. **Regulation:** clinician decision support, built for DPDP consent (consent-manager duties from 13 Nov 2026) and the CDSCO software pathway.

The ask: one diabetes clinic, 30 consenting adults with Type 2 diabetes on CGM for 12 weeks, and a doctor willing to review the twin's forecasts.

Pilot design and targets are proposals; outcomes will be reported as associations, not causal proof. ABHA figures: PIB fact sheet, 6 Jul 2026. Regulatory dates: DPDP Rules (notified 13 Nov 2025); CDSCO draft guidance on medical device software (21 Oct 2025).